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Tyrone Ghaswala

Curriculum Vitae

Appointments

2024–present **Assistant Professor Teaching Stream**, *Centre for Education in Mathematics and Computing, University of Waterloo*, Waterloo, Canada

2022–present **Adjunct Professor**, *Department of Pure Mathematics, University of Waterloo*, Waterloo, Canada

2022–2024 **Lecturer**, *Centre for Education in Mathematics and Computing, University of Waterloo*, Waterloo, Canada

2020–2022 **CIRGET Postdoctoral Fellow**, *Université du Québec à Montréal*, Montréal, Canada

2018–2019 **PIMS Postdoctoral Fellow**, *University of Manitoba*, Winnipeg, Canada

Education

2012–2017 **PhD, Pure Mathematics**, *University of Waterloo*, Waterloo, Canada
The liftable mapping class group

2011–2012 **MMath, Pure Mathematics**, *University of Waterloo*, Waterloo, Canada
Intersection Forms on Fermat Hypersurfaces in $\mathbb{CP}^3$

2008–2010 **BSc, Mathematical Physics**, *University of Melbourne*, Parkville, Australia

Papers and Preprints

10. **Rigidity of braid group actions on $\mathbb{R}$ and of low-genus mapping class group actions on S^1**, *w Idrissa Ba and Adam Clay*, To appear in *Annales de l'institut Fourier*
9. **Cofinal elements and fractional Dehn twist coefficients**, *w Adam Clay*, *International Mathematics Research Notices*, 2024-9 (2023) 7401-7420
8. **Circularly ordering direct products and the obstruction to left-orderability**, *w Adam Clay*, *Pacific Journal of Mathematics*, 312-2 (2021) 401-419
7. **Homeomorphic subsurfaces and the omnipresent arcs**, *w Federica Fanoni and Alan McLeay*, *Annales Henri Lebesgue*, 4 (2021) 1565-1593
6. **Promoting circular-orderability to left-orderability**, *w Jason Bell and Adam Clay*, *Annales de l'institut Fourier*, 71(1) (2021) 175-201
5. **Big Torelli groups: generation and commensurations**, *w Javier Aramayona, Autumn E. Kent, Alan McLeay, Jing Tao, and Rebecca R. Winarski*, *Groups, Geometry, and Dynamics*, 13(4) (2019) 1373-1399
4. **Free products of circularly ordered groups with amalgamated subgroup**, *w Adam Clay*, *Journal of the London Mathematical Society* (2), 100 (2019) 775-803
3. **Mapping class groups of covers with boundary and braid group embeddings**, *w Alan McLeay*, *Algebraic & Geometric Topology*, 20 (2020) 239-278

2. **Lifting homeomorphisms and cyclic branched covers of spheres**, *w Rebecca R. Winarski*, *Michigan Mathematical Journal*, 66(4) (2017) 885-890
1. **The liftable mapping class group of balanced superelliptic covers**, *w Rebecca R. Winarski*, *New York Journal of Mathematics*, 23 (2017) 133-164
0. **The liftable mapping class group**, *PhD thesis*, Tyrone Ghaswala (2017). UWSpace. http://hdl.handle.net/10012/12075

Talks

Invited Conference Talks

Dec 2025	CMS Winter Meeting (Topology session), Toronto
Dec 2025	CMS Winter Meeting (An invitation to low-dimensional topology session), Toronto
June 2023	(Plenary) Nearly Carbon Neutral Geometric Topology Conference, Virtual conference
May 2023	Workshop on Orderable Groups, Centre de recherches mathématique, Montréal
Dec 2022	Canadian Mathematical Society Winter Meeting, Toronto
Apr 2022	Graduate Student Topology and Geometry Conference, Georgia Institute of Technology
Dec 2021	Canadian Mathematical Society Winter Meeting, Virtual conference
June 2021	Nearly Carbon Neutral Geometric Topology Conference, Virtual conference
June 2021	Canadian Mathematical Society Summer Meeting, Virtual conference
June 2020	Nearly Carbon Neutral Geometric Topology Conference, Virtual conference
Dec 2019	Canadian Mathematical Society Winter Meeting, Toronto
Sep 2019	Low-dimensional topology program, Université du Québec à Montréal
June 2019	Ordered Groups and Rigidity in Dynamics and Topology, Casa Matemática Oaxaca, Oaxaca
June 2019	Canadian Mathematical Society Summer Meeting, Regina, SK
May 2019	Surfaces of Infinite Type, American Institute of Mathematics, San Jose (short talk)
Dec 2018	Canadian Mathematical Society Winter Meeting, Vancouver
Sep 2017	American Mathematical Society Fall Eastern Sectional Meeting, SUNY Buffalo
May 2017	Braids in Algebra, Geometry and Topology, Edinburgh (lightning talk)
Dec 2016	Canadian Mathematical Society Winter Meeting, Niagara Falls
June 2016	Topology Students Workshop, Georgia Institute of Technology
Mar 2016	Spring Topology and Dynamics Conference, Baylor University

Invited Seminar and Colloquium Talks

Feb 2026	CIRGET Geometry and Topology Seminar, Université du Québec à Montréal
Nov 2025	Geometry and Topology Seminar, McMaster University
May 2023	UTSC CMS Undergraduate Seminar, University of Toronto Scarborough (virtual)
May 2023	Phillips Exeter Academy math seminar, Phillips Exeter Academy
Mar 2023	Hyperbolic Lunch, University of Toronto
Feb 2023	Geometry and Topology Seminar, McMaster University
Feb 2023	Geometric Topology Seminar, Columbia University
Feb 2023	Columbia Undergraduate Mathematics Colloquium, Columbia University
Nov 2022	Geometry and Topology Seminar, University of Waterloo
Sep 2022	Topology seminar, University of Wisconsin-Milwaukee (virtual)
Apr 2022	SAG Seminar, Université de Sherbrooke
Mar 2022	CIRGET Geometry and Topology Seminar, Université du Québec à Montréal

Mar 2022	Hyperbolic Geometry Seminar, CUNY (virtual)
Nov 2021	Geometric Group Theory Seminar, McGill University
Mar 2021	Australian Geometric Topology Webinar
Mar 2021	Geometry and Topology Seminar, Georgia Tech
Sep 2020	Max Dehn Seminar, University of Utah
Jul 2020	Online Geometry Seminar, ETH Zurich
Feb 2020	Geometric Group Theory Seminar, McGill University
Jan 2020	CIRGET Geometry and Topology Seminar, Université du Québec à Montréal
Sep 2019	Geometric Topology Seminar, Columbia University
Oct 2018	Geometry and Topology Seminar, University of Melbourne
Oct 2018	Topology Seminar, Monash University
Sep 2018	Geometry and Topology Seminar, University of Waterloo
Sep 2018	Geometry and Topology Seminar, McMaster University
Sep 2018	Algebra Seminar, University of Waterloo
Sep 2018	Department of Mathematics and Statistics Colloquium, University of Calgary
Mar 2018	Geometry and Topology Seminar, Brown University
Mar 2018	Geometric Group Theory and Topology Seminar, Tufts University
Dec 2017	Geometry and Topology seminar, McMaster University
Sep 2017	Topology Seminar, University of Michigan
May 2017	Geometry and Topology Seminar, University of Manchester
Mar 2017	Geometric Group Theory Seminar, Ohio State University
Feb 2017	Geometry and Topology Seminar, McMaster University
Jan 2017	Geometry and Topology Seminar, University of Waterloo
May 2016	Joint Pure Math and C&O Grad Student Colloquium, University of Waterloo
Apr 2016	Groups and Combinatorics Seminar, University of Western Australia
Apr 2016	Geometry and Topology Seminar, University of Sydney
Apr 2016	Geometry and Topology Seminar, University of Melbourne
Apr 2016	CARMA Colloquium, University of Newcastle
Mar 2016	Topology Seminar, Monash University
Mar 2016	Algebra Seminar, MacMaster University
Feb 2016	Algebra Seminar, University of Waterloo
Feb 2015	Pure Math Graduate Student Colloquium, University of Waterloo
Sep 2012	Pure Math Graduate Student Colloquium, University of Waterloo

Supervision

2018 USRA	**Robert Harris**, *Density of representations of the free group*, (co-supervisor)
2024 USRA	**Matteus Froese**, *Rigidity of surface braid groups*
2024 USRA	**Elysia Wang**, *Dehn Nielsen Baer for bounded surfaces*
2025 USRA	**Justin Nagarajah**, *Counting circular orderings*
2026 USRA	**Mason Walton**, *Self covers of infinite-type surfaces*

Service

Conferences and seminars co-organised

2025	Ontario Topology Seminar, Western University
2022	CIRGET Low-dimensional topology working group, Université du Québec à Montréal
2020-2021	Big Surf(aces) virtual seminar
2020-2021	CIRGET Low-dimensional topology working group, Université du Québec à Montréal
2019	63rd Cascade Topology Seminar, University of Manitoba
2018	Geometry and topology learning seminar, University of Manitoba
2017	Low-dimensional topology and geometric group theory, CMS Winter meeting special session, University of Waterloo
2014–2015	Group Presentations learning seminar, Unviersity of Waterloo
2013	Group Cohomology learning seminar, University of Waterloo
2013	Morse Theory learning seminar, University of Waterloo
2013–2015	Joint Pure Mathematics and Combinatorics and Optimization Graduate Student Colloquium, University of Waterloo

Mathematical outreach

Sep 2017	TEDxUofTSalon, Fields Institute, University of Toronto
Apr 2017	Fields Thesis Competition, University of Toronto (3rd Place)
Mar 2017	University of Waterloo Three Minute Thesis finals
Apr 2016	Lunchmaths Seminar, Monash University
2012–2016	Math Circles Presenter, CEMC, University of Waterloo
June 2014	Mathematics Union Seminar, University of Western Australia
2010,'11,'13,'23	Australian National Mathematics Summer School
May 2011	Monday Evening Program, Queen's College, University of Melbourne
2008 – present	Various high schools across Australia and Canada

Referee for

Adv. Math., Algebr. Geom. Topol., Bol. Soc. Mat. Mexicana., Canad. J. Math., Groups Geom. Dyn., Houston J. Math., Internat. J. Algebra Comput., Int. Math. Res. Not., Michigan Math. J., New York J. Math., Proc. Amer. Math. Soc., Proc. Cambridge Philos. Soc., Topology Appl.

Reviewer for

National Science Centre, Poland.

Other

2020–	Contest Problem Creater, CEMC, University of Waterloo
2014–2016	Graduate Student Representative, Pure Mathematics departmental meetings
2012–2016	Math Contest Marker, CEMC, University of Waterloo
2014–2015	Chief Returning Officer, University of Waterloo Graduate Student Association
2013–2015	Graduate Student Representative, University Committee on Student Appeals, University of Waterloo
2012–2015	Pure Mathematics Councillor, University of Waterloo Graduate Student Association Council
2012–2015	Activities Committee and Finance Committee member, University of Waterloo Graduate Student Association
2013–2014	Deputy Returning Officer, University of Waterloo Graduate Student Association
2013–2014	Pure Math Representative, Graduate Student Computing Advisory Committee, University of Waterloo
2012–2014	Math Faculty Graduate Student Representative, Senate Graduate and Research Council, University of Waterloo

2010	Treasurer, Intercollegiate Council, University of Melbourne residential colleges
2010	Arts Representative, Queen's College Sports and Social Club general committee
2009–2010	Victorian State Representative, National Association of Australian University Colleges
2009	Treasurer, Intercollegiate Activities Committee, University of Melbourne residential colleges

Conferences Attended

Dec 2025	Canadian Mathematical Society Winter Meeting, Toronto, ON
Oct 2025	Ontario Topology Seminar, London, ON
June 2025	Blooming Beasts: A Conference on the Geometry, Topology, and Dynamics of Infinite-Type Surfaces, Casa Matemática Oaxaca, Oaxaca, MX
Aug 2024	What's your trick?, BIRS, Banff, AB
June 2024	Topology Students Workshop (as a mentor), Nashville TN
June 2023	Nearly Carbon Neutral Geometric Topology Conference, Virtual conference
May 2023	Workshop on Orderable Groups, Centre de recherches mathématique, Montréal, QC
Dec 2022	Canadian Mathematical Society Winter Meeting, Toronto, ON
Oct 2022	Big Mapping Class Groups and Diffeomorphism Groups, CIRM, Luminy, France
Aug 2022	Communicating Mathematics, Cornell University, Ithaca, NY
June 2022	Geometry, Arithmetic, and Groups, UT Austin, Austin, TX
Apr 2022	Graduate Student Topology and Geometry Conference, Georgia Institute of Technology, Atlanta, GA
Aug 2021	Summer Surfacing, Virtual conference
Jul 2021	Tech Topology summer school, Virtual conference
June 2021	Nearly Carbon Neutral Geometric Topology Conference, Virtual conference
June 2021	Canadian Mathematical Society Summer Meeting, Virtual conference
Jul 2020	Topology Postdocs Workshop, Virtual conference
June 2020	Nearly Carbon Neutral Geometric Topology Conference, Virtual conference
Dec 2019	Canadian Mathematical Society Winter Meeting, Toronto, ON
Sep 2019	Program in low-dimensional topology, Université du Québec à Montréal, Montréal, QC
Sep 2019	63rd Cascade Topology Seminar, University of Manitoba, Winnipeg, MB
June 2019	Ordered Groups and Rigidity in Dynamics and Topology, Casa Matemática Oaxaca, Oaxaca, MX
June 2019	Canadian Mathematical Society Summer Meeting, Regina, SK
May 2019	Surfaces of Infinite Type, American Institute of Mathematics, San Jose, CA
Dec 2018	Canadian Mathematical Society Winter Meeting, Vancouver, BC
Apr 2018	Geometry of Teichmüller space and mapping class groups, University of Warwick, Coventry, UK
Mar 2018	First Canadian Geometry and Topology Seminar, Fields Institute, Toronto, ON
Dec 2017	Canadian Mathematical Society Winter Meeting, Waterloo, ON
Sep 2017	American Mathematical Society Fall Eastern Sectional Meeting, SUNY Buffalo, NY
May 2017	Braids in Algebra, Geometry and Topology, International Centre for Mathematical Sciences, Edinburgh, UK
Jan 2017	Joint Mathematics Meeting, Atlanta, GA
Dec 2016	Canadian Mathematical Society Winter Meeting, Niagara Falls, ON
Nov 2016	Virginia Topology Conference, University of Virginia, Charlottesville, VA
June 2016	Topology Students Workshop, Georgia Institute of Technology, Atlanta, GA
Mar 2016	Spring Topology and Dynamics Conference, Baylor University, Waco, TX

Dec 2015	Canadian Mathematical Society Winter Meeting, Montreal, ON
June 2015	Summer Graduate School in Geometric Group Theory, Mathematical Sciences Research Institute, Berkeley, CA
May 2015	Workshop on Mapping Class Groups and Out(F_n), University of Austin at Texas, Austin, TX
Dec 2014	Canadian Mathematical Society Winter Meeting, Hamilton, ON
Jan 2014	Joint Mathematics Meeting, Baltimore, MD
Dec 2013	Canadian Mathematical Society Winter Meeting, Ottawa, ON
Sep 2012	Topology of Algebraic Varieties, Centre de Recherche Mathématiques, Montreal, QC
May 2012	Geometry and Physics, University of Waterloo & Perimeter Institute, Waterloo, ON

Teaching

Awards

2016 **Outstanding Graduate Teaching Award**, *Pure Mathematics Department, University of Waterloo*
For outstanding teaching by a Graduate Student as both a lecturer and a teaching assistant.

2014 **Teaching Assistant Award**, *Pure Mathematics Department, University of Waterloo*
For outstanding teaching by a Graduate Student in a teaching assistant role.

Courses

Winter 2026	MATH 648 - Foundations of Calculus II, University of Waterloo (online)
Fall 2025	MATH 647 - Foundations of Calculus I, University of Waterloo (online)
Spring 2025	PMATH 321 - Non-Euclidean Geometry, University of Waterloo, University of Waterloo
Spring 2025	PMATH 336 - Introduction to Group Theory with Applications, University of Waterloo
Spring 2025	MATH 661 - Geometry, University of Waterloo
Winter 2025	MATH 237 - Calculus 3 for Honours Mathematics, University of Waterloo
Winter 2025	PMATH 340 - Elementary Number Theory, University of Waterloo
Spring 2023	PMATH 340 - Elementary Number Theory, University of Waterloo
Fall 2022	MATH 235 - Linear Algebra 2 for Honours Mathematics, University of Waterloo
Fall 2022	MATH 135 - Algebra for Honours Mathematics, University of Waterloo
Winter 2021	MATH 141 - Calculus 2, McGill University (online)
Fall 2020	MATH 140 - Calculus 1, McGill University (online)
Winter 2020	MATH 133 - Linear Algebra and Geometry, McGill University
Winter 2019	MATH 1240 - Elementary Discrete Mathematics, University of Manitoba
Summer 2018	MATH 2090 - Linear Algebra 2, University of Manitoba
Winter 2018	MATH 1210 - Techniques of Classical and Linear Algebra, University of Manitoba
Fall 2017	MATH 135 - Algebra for Honours Mathematics, University of Waterloo
Fall 2016	MATH 225 - Applied Linear Algebra 2, University of Waterloo
Fall 2015	PMATH 336 - Introduction to Group Theory with Applications, University of Waterloo

Other

2012–2017 **Online Teaching Assistant**, *University of Waterloo*

2011–2017 **Teaching Assistant**, *University of Waterloo*

2009–2011 **Mathematics Tutor**, *Various residential colleges, University of Melbourne*

Non-Academic Experience

2013–2015 **Events Coordinator**, *Graduate House, University of Waterloo*
Events coordinator at the on campus graduate pub, the Graduate House.

2011 **Mentor**, *Queen's College, University of Melbourne*
Responsible for mentoring a group of 10 first year undergraduate students making the transition into University life on campus.

2009 **Producer - Cabaret**, *Queen's College Music and Drama Society*
Produced the Queen's College production of Cabaret, overseeing a cast and crew of over 100 people and a $20,000 budget.

Awards and Honours

Academic

2016 **25th Anniversary FEZANA Endowment Scholarship for Excellence**, *Federation of Zoroastrian Associations of North America*
For academic excellence.

2008–2010 **6 time Wyvern Medalist**, *Queen's College, University of Melbourne*
For maintaining an average above 80% for the semester.

2010 **Master's Scholarship**, *Queen's College, University of Melbourne*
For oustanding academic achievement.

2009 **Johnstone-Need Major Scholarship**, *Queen's College, University of Melbourne*
For outstanding academic achievement.

2008 **Melbourne National Scholarship**, *University of Melbourne*
An undergraduate entrance scholarship for academic achievement.

2008 **A.L.Blakers Scholar**, *Australian National Mathematics Summer School*
For the largest social and academic contribution to the National Mathematics Summer School by a student.

Extracurricular

2012, 2014 **Intramural Squash Champion**, *University of Waterloo*
2-time winner of the intramural squash competition.

2012 **Volunteer of the Year**, *Graduate Student Association, University of Waterloo*

2009 **Murray Sutherland award**, *Queen's College, University of Melbourne*
For outstanding contribution to the arts at Queen's College.

Interests

Music Folk and jazz guitar, casual pianist, shower vocalist

Puzzles Principal Enigmatologist for Key Clues (2022 - present), Past puzzle writer for the CMS student committee's publication *Notes from the Margin*, puzzle writer for the 2015 MUMS Puzzle Hunt, organiser of the 2016 MUMS Puzzle Hunt, and member of a team competing in international puzzle hunt competitions.

Sport Squash, cycling, basketball, wrestling with my toddlers to get them to go to sleep.

References

Avaliable upon request.